



BITS Pilani presentation

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Pilani Campus

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SS ZG622

Software Project Management

Lecture No. 2

Objectives, buy vs build & process model selection



1. Project vs. project management objectives
2. Build versus buy decision
3. Process model selection
4. Plan driven process models
5. Agile methods

What is a Project?

- A complex, non-routine, one-time effort limited by time, budget, resources, and performance specifications designed to meet customer needs.
- A project is unique, has specific time frame (not a continuing or ongoing activity like automobile production, ...), requires resources, interdependent activities, and has a degree of uncertainty.

Project Stakeholders

These are people who have a stake or interest in the project

In general, they could be *users/clients* or *developers/implementers*

They could be:

- Within the project team
- Outside the project team, but within the same organization
- Outside both the project team and the organization (society,...)

Different stakeholders may have different objectives – need to define common project objectives

Key Stakeholders



Project characteristics



- Risks and uncertainties e.g.
 - are requirement well understood?
 - are technologies to be used well understood?
- Type of application being built e.g.
 - information system or embedded system?
 - criticality vs. differences between target and development environments?
- Clients' own requirements
 - need to use a particular method

Activity W02#1



Differentiate project objectives and project management objectives for ACL Mux Core Project.

Project objectives vs

Project management objectives



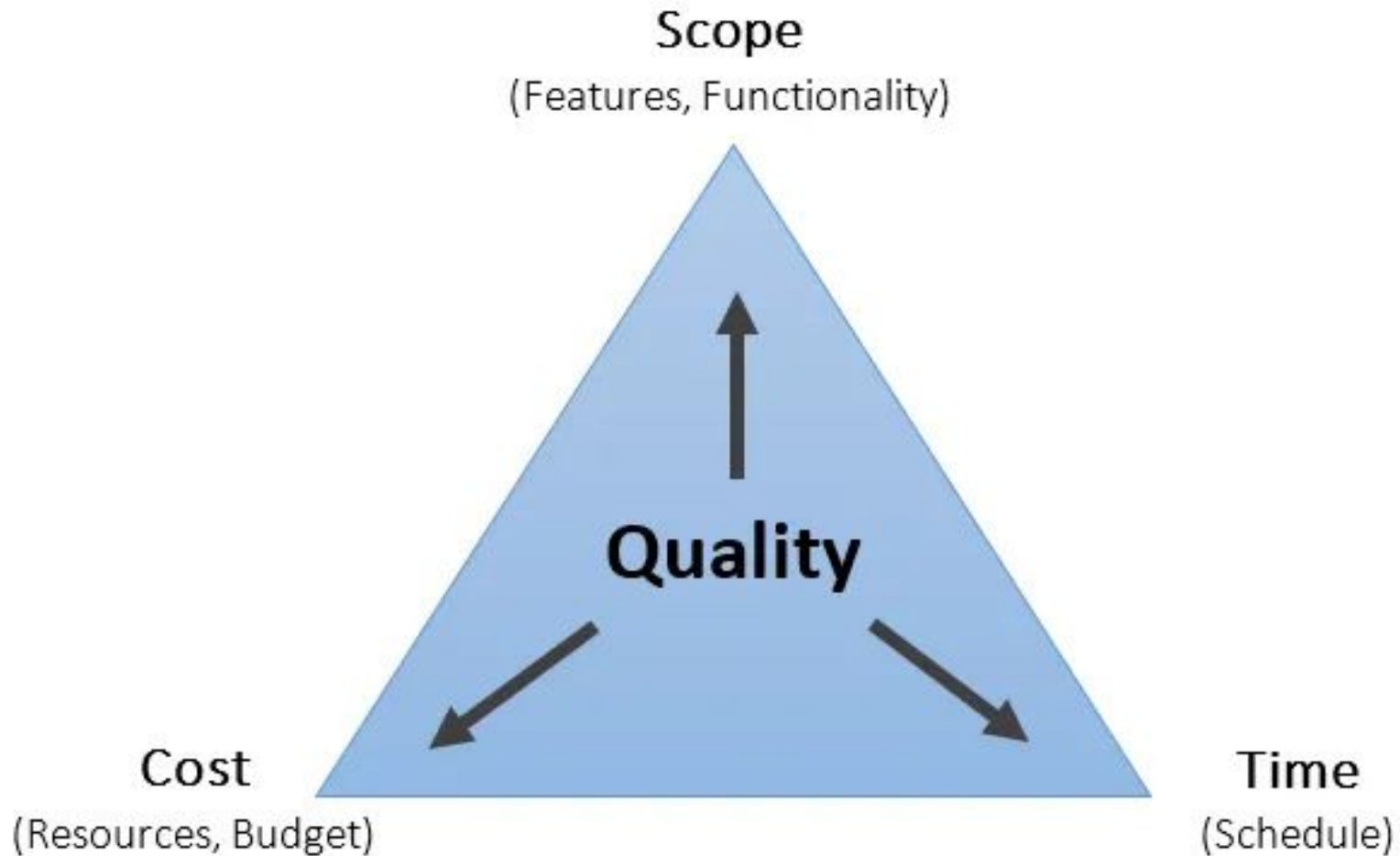
Project objectives

- define the **desired outcomes or deliverables**, focusing on the **end product or system** and its alignment with business goals.
- describe **what** the project aims to achieve in terms of functionality, performance, and value to the organization.
- centered on the **product** and its impact on stakeholders, such as customers, employees, or the business.

PM objectives

- focus on the **process of delivering the project**, emphasizing how the project is planned, executed, and controlled to ensure successful completion.
- describe **how** the project will be managed to achieve the project objectives.
- centered on the **project execution**, including timelines, budgets, resources, risks, and stakeholder coordination.

Project management triangle



<https://medium.com/@harpreet.dhillon/iron-triangle-triple-constraints-of-project-management-e818e631826c>

Build vs Buy Decision



- Option 1 - Buy a COTS solution
- Option 2 - Build in-house
- Option 3 - Build with a third party vendor

Option 1: Buy a commercial off the shelf (COTS) solution



- Purchase a pre-built software solution from a vendor that meets the project's functional requirements (e.g., CRM, ERP, or industry-specific systems). Implementation involves licensing, configuring the software to fit organizational needs, and integrating it with existing systems. Vendors typically provide cloud-based or on-premise options, along with ongoing support and updates.
- **Key Benefits:** Fast deployment (3-6 months), lower initial costs, and reduced development risk due to proven solutions. Ideal for standard requirements and organizations with limited technical resources.
- **Key Drawbacks:** Limited customization, potentially requiring workarounds for unique needs. Ongoing subscription or maintenance fees and dependency on vendor support.

Option 2: Build in-house



- Develop a custom software solution using the organization's internal IT team, tailored to specific requirements. The process involves defining requirements, designing, coding, testing, and deploying the system, following a software development life cycle (SDLC). The organization handles ongoing maintenance and updates.
- **Key Benefits:** High customization to meet unique needs, full control over the system, and no recurring licensing fees. Enables innovative features and aligns closely with organizational goals.
- **Key Drawbacks:** High costs, long timelines (12-24 months), and significant risk due to complexity and resource demands. Requires a skilled in-house team and robust project management.

Option 3: Build with a third-party vendor



- Partner with a software development vendor to build a custom solution tailored to the organization's needs. The vendor manages development, testing, and integration using methodologies like Agile, while the organization provides requirements and oversight. Maintenance can be vendor-managed or transitioned in-house.
- **Key Benefits:** Balances customization with professional expertise, moderate costs, and faster delivery (9-18 months) than in-house builds. Reduces risks through vendor experience and resources.
- **Key Drawbacks:** Higher costs than COTS, some vendor dependency, and requires strong coordination to align requirements. Potential for delays if scope changes occur.

Activity W02#2



Which one of the three options is best suited for the Mux Core System at ACL? And, why?

Development approaches



Plan-driven approaches

- Step-by-step methods where each step and intermediate product is carefully defined
- Emphasis on getting quality right first time
- Rapid application development is an alternative

Agile approaches

- Emphasis on speed of incremental delivery rather than documentation

Process model choices

- Waterfall model
- Evolutionary process model
- Incremental and iterative process models (e.g., Unified process)

Or

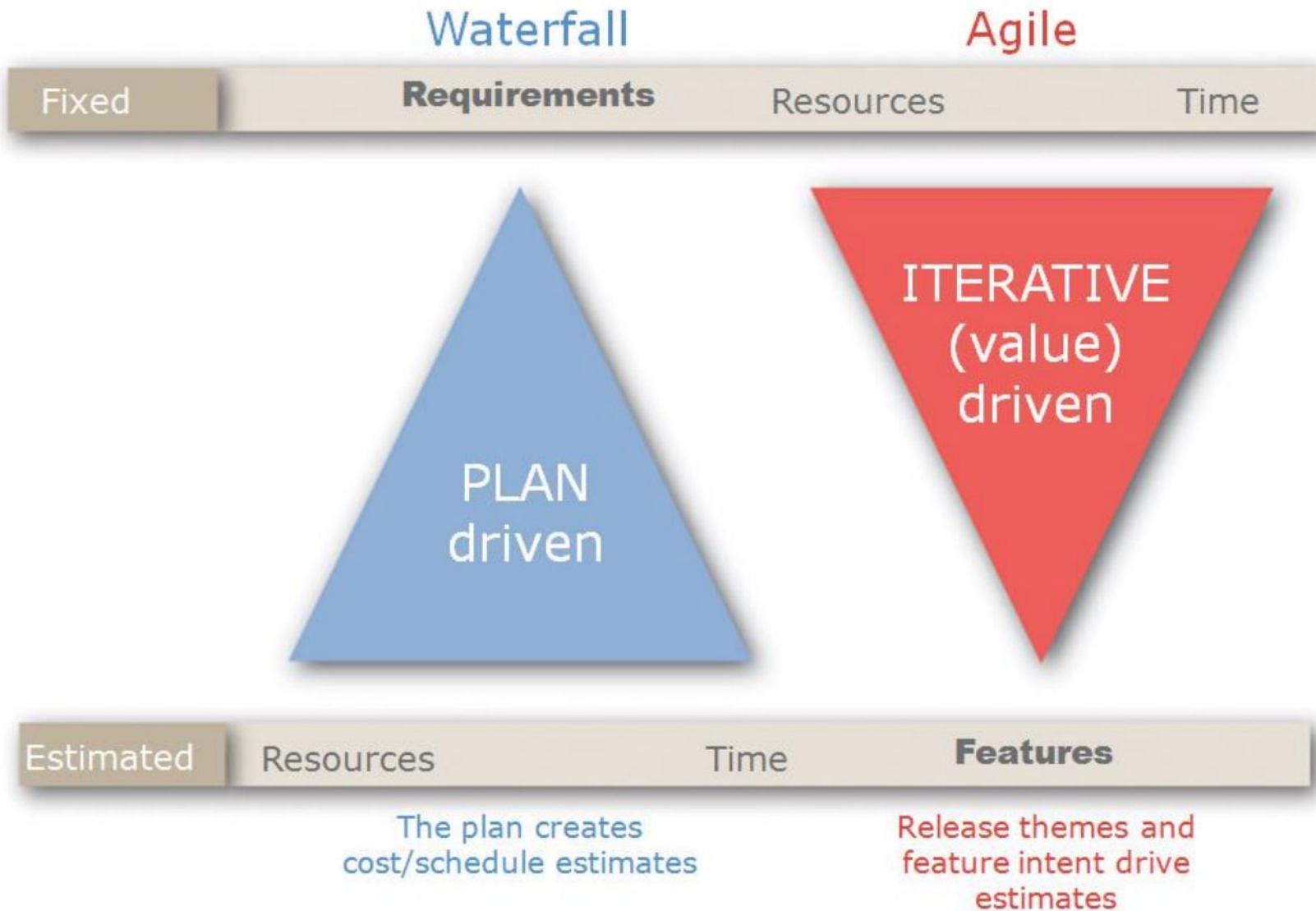
- Agile methods (e.g., Scrum, XP, ...)

Plan-driven process models vs. agile methods



- In plan-driven process models, activities are planned in advance and progress is measured against the plan
- In agile process models, planning is incremental and it is easier to change the process to reflect changing requirements
- There is no right or wrong process model; most organizations/teams include elements from both in their process models

The Agile Paradigm Shift



Activity W02#3



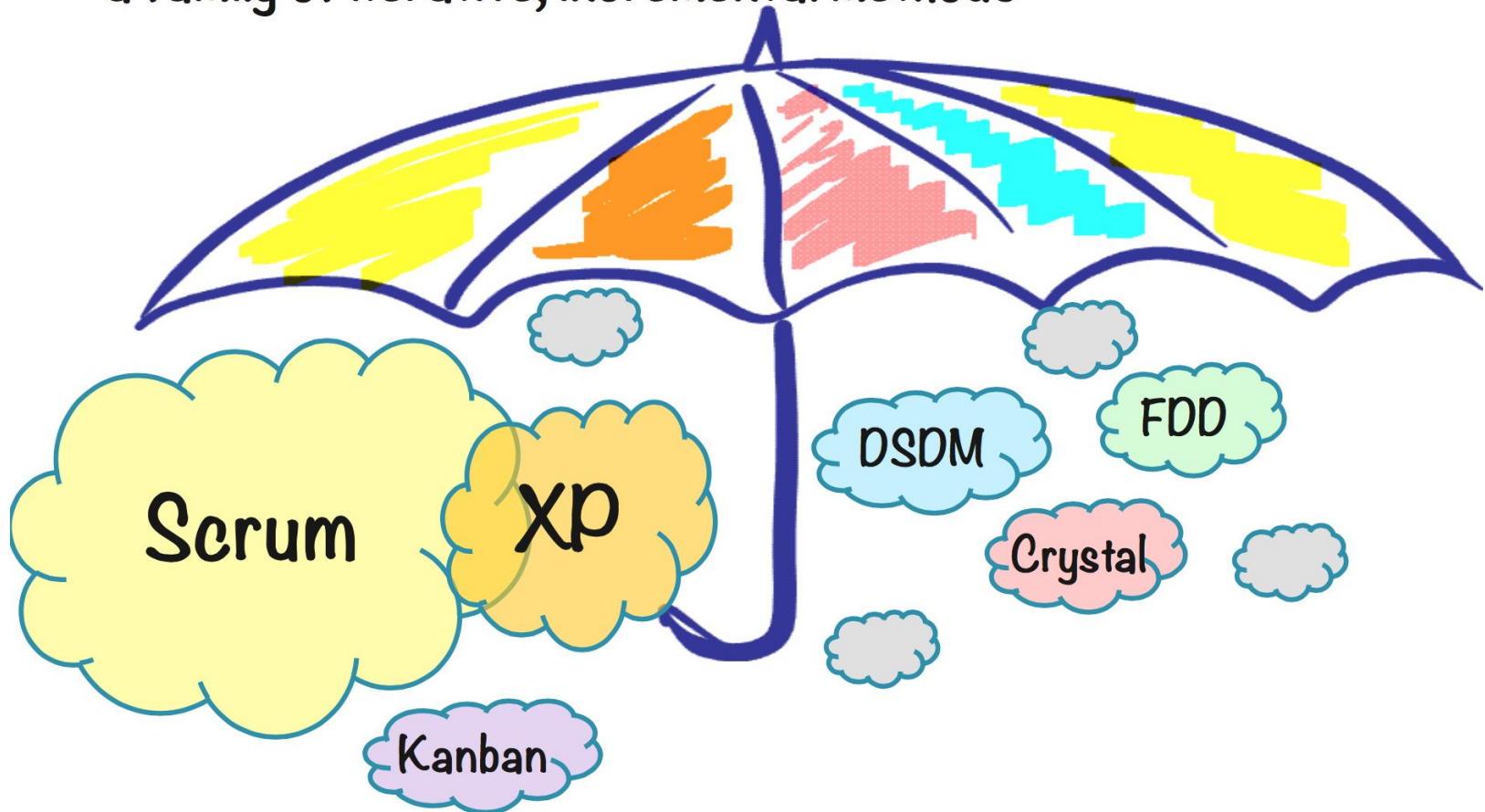
Identify pros and cons of using the following process models for the Mux-Core Project

- waterfall
- prototyping
- iterative/incremental

Agile methods



Agile "umbrella" –
a family of iterative, incremental methods

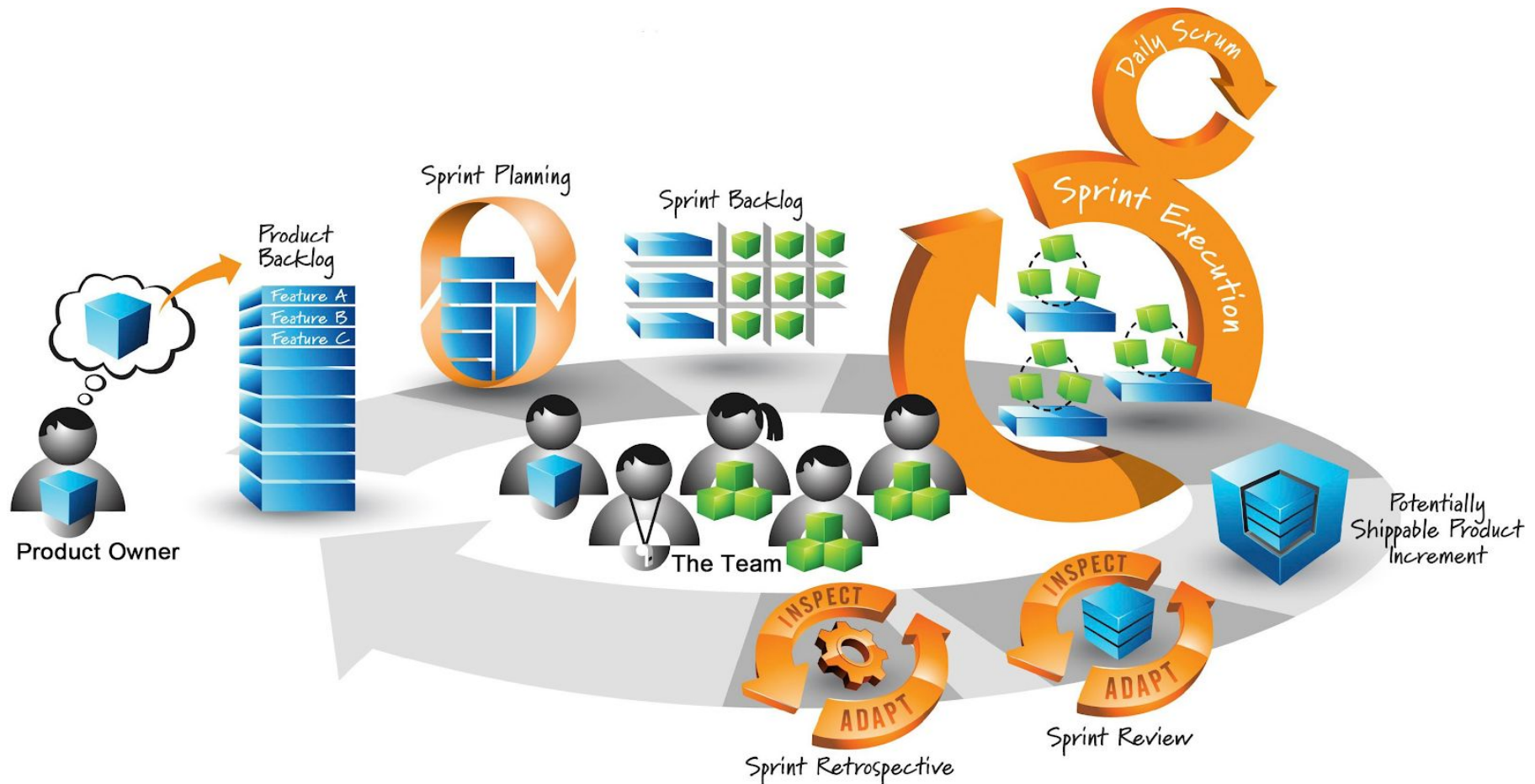




Scrum overview

- A framework within the agile paradigm that organizes work into time-boxed iterations called sprints (typically 2-4 weeks).
- Emphasizes roles (Product Owner, Scrum Master, Development Team), artifacts (Product Backlog, Sprint Backlog, Increment), and events (Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective).
- Focuses on project management and team collaboration, providing a structure for delivering increments of working software while adapting to changing requirements.

Agile method Scrum



<http://www.programmeronrails.com/2016/03/20/scrum-overview/>



Extreme Programming Overview

- An agile methodology that focuses on engineering practices to improve software quality and responsiveness to customer needs.
- Emphasizes frequent releases, continuous feedback, and technical excellence through practices like pair programming, test-driven development (TDD), continuous integration, and refactoring.
- More prescriptive about how development work is performed compared to Scrum.
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Activity W02#4



- a) Do you think Scrum is the right process model for the Mux-Core Project development?
- b) How XP can be used during the development?

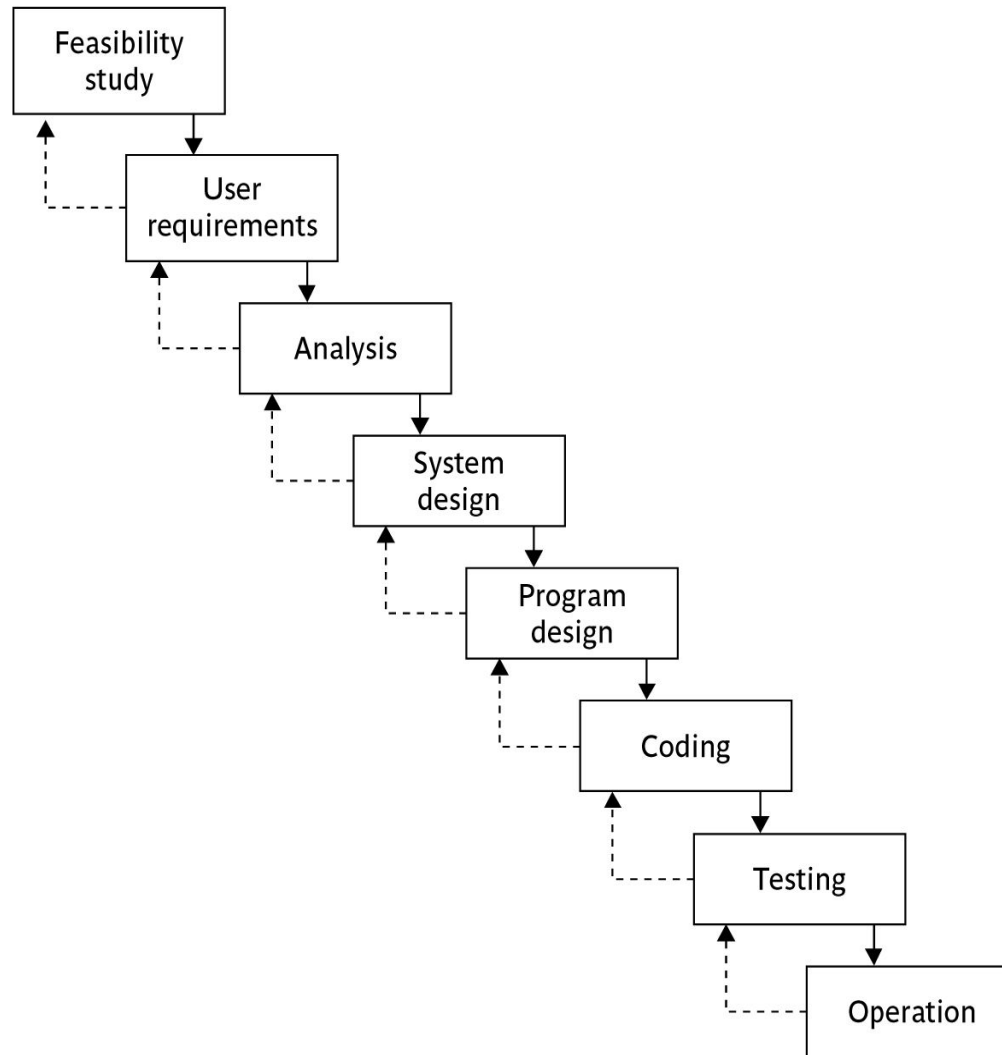
Activity W02#5



Explain how Scrum is different from the iterative and incremental process model.

Additional slides (on process models)

Waterfall Model



Waterfall

- the 'classical' model
- imposes structure on the project
- every stage needs to be checked and signed off
- BUT
 - limited scope for iteration
- V model approach is an extension of waterfall where different testing phases are identified which check the quality of different development phases

Evolutionary prototyping

An iterative process of creating quickly and inexpensively live and working models to test out requirements and assumptions'

- 'throw away' prototypes
- evolutionary prototypes

what is being prototyped?

- human-computer interface
- functionality

Reasons for prototyping

- learning by doing
- improved communication
- improved user involvement
- a feedback loop is established
- reduces the need for documentation
- reduces maintenance costs i.e. changes after the application goes live
- prototype can be used for producing expected results

Prototyping: some dangers

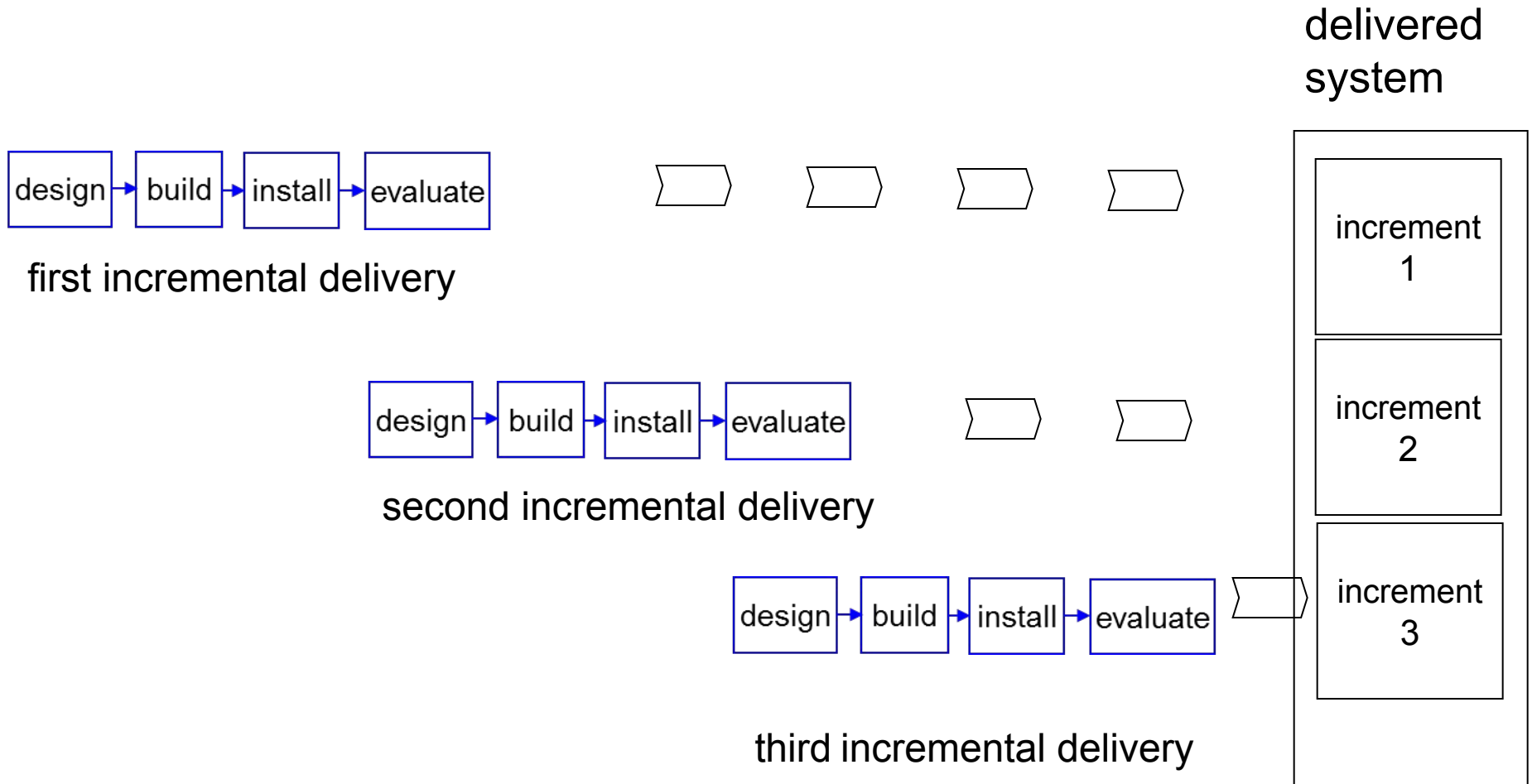
- users may misunderstand the role of the prototype
- lack of project control and standards possible
- additional expense of building prototype
- focus on user-friendly interface could be at expense of machine efficiency

Categories of Prototypes



- Throw-away Prototypes
 - ‘experimental’ (HW/SW prototype)
 - ‘exploratory’ (application prototype)
- Re-usable Prototype
 - ‘evolutionary’ build (for continuous feedback)
 - Final prototype = Deliverable!

Incremental and iterative models



Incremental and iterative models: benefits



- feedback from early stages used in developing latter stages
- shorter development thresholds
- user gets some benefits earlier
- project may be put aside temporarily
- reduces 'gold-plating':

BUT there are some possible disadvantages

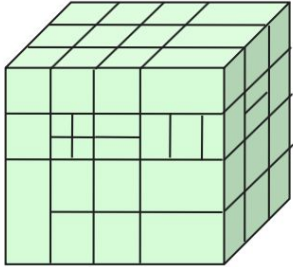
- loss of economy of scale
- 'software breakage'

Incremental planning

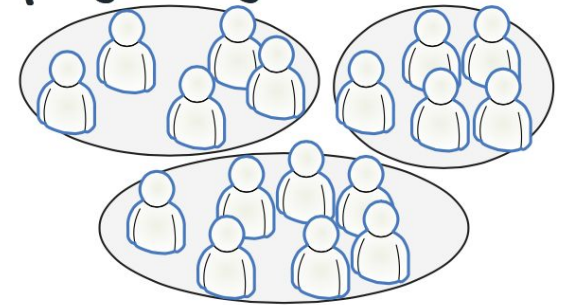
- steps ideally 1% to 5% of the total project
- ideal if a step takes one month or less:
 - not more than three months
- each step should deliver some benefit to the user
- some steps will be physically dependent on others

Scrum in a nutshell

Split your product

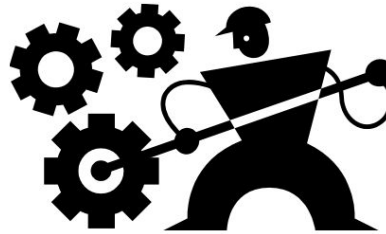


Split your organization

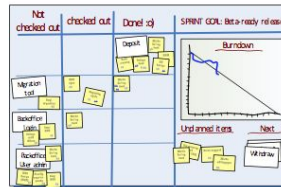
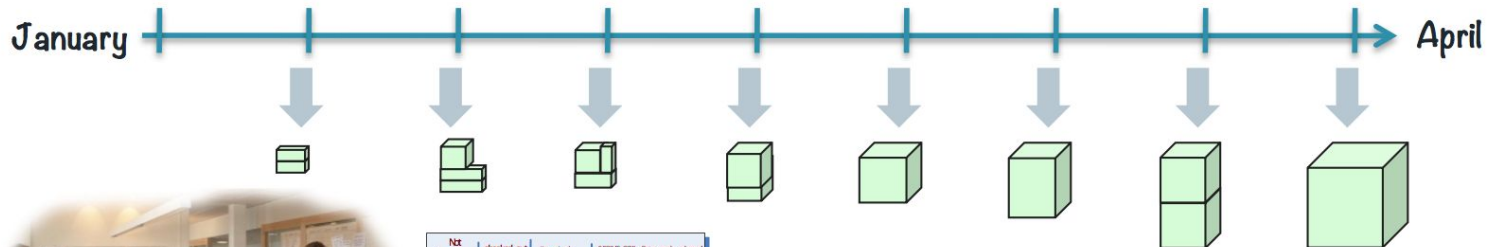


~~Large group~~ spending ~~a long time~~ building a ~~huge thing~~
~~Small team~~ spending a little time building a small thing
... but integrating regularly to see the whole

Optimize process



Split time



Scrum overview – structure

